

Traps, Vents and Sewer Gas

Two of your most repeated misses are here, and both are the same mistake: answering "everywhere" for a device that solves one specific problem.

A trap holds a plug of water that sewer gas cannot pass. Every fixture has one. The problem is that draining water drags air with it, and that suction will pull the water straight out of the trap unless air can get in behind it. That is what the vent system is for.

The numbers here

P-trap water seal	2 to 4 in
Plumbing vents above the roof	6 in

Vent stack

The vertical upper portion above the topmost fixture, through which gases and odors escape the sanitary drainage system.

Wet vent

A vent serving fixtures from as many as two bathrooms on the same floor.

YOU HAVE MISSED THIS

The purpose of vent pipes in a drainage system is to:
Ensure the rapid flow of waste.

YOU HAVE MISSED THIS 2 TIMES

The absence of vent stacks indicates that the:
Plumbing code has been violated.

YOU HAVE MISSED THIS

Plumbing vent pipes cannot be _____ in size.
Decreased.

A vent may be enlarged but never reduced along its run, because a narrowing throttles the airflow the whole system depends on.

The two devices you keep answering "at all traps"

YOU HAVE MISSED THIS 4 TIMES

Where is an air admittance valve used?
In a kitchen island for the sink.

YOU HAVE MISSED THIS 3 TIMES

Where is a trap primer used?

At an emergency floor drain for a washing machine.

Between them you have missed these seven times, and both times the pull is to answer that they go at all traps. Neither is general. Each solves one specific situation that ordinary plumbing cannot:

Air admittance valve	Opens on negative pressure to admit air, then closes to block sewer gas. It exists for a fixture with no path to a roof vent — an island sink in the middle of a kitchen floor is the case, because there is no wall to run a vent up.
Trap primer	Feeds a trickle of water to a trap that would otherwise dry out. It exists for a drain nobody uses — an emergency floor drain by a washing machine. A dry trap is an open pipe to the sewer.

Ask what problem the device solves and the answer places itself. No vent path means an air admittance valve. No water going down the drain means a trap primer. A trap primer also requires an air gap, so that the drain can never siphon back into the supply feeding it.

Say it back

Cover the page. Answer each one out loud.

1. What does a trap hold, and what tries to remove it?
2. P-trap seal depth, and how far vents rise above the roof?
3. What is a vent stack, and what is a wet vent?
4. State the purpose of vent pipes in one phrase.
5. May a vent be reduced in size along its run?
6. Where does an air admittance valve go, and what problem does it solve?
7. Where does a trap primer go, what problem does it solve, and what does it require?